

Tanishka Gupta

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Education

Graphic Era (Deemed to be) University — B.Tech in Computer Science & Engineering (AI-ML) | 2024 – 2028 | CGPA: 9.02/10

Projects

- **Hotel Management System (C):** Developed a modular C-based hotel management application with room booking, customer record handling, billing features, and persistent data storage using file handling.
- **Expense Tracker (C):** Built a command-line expense tracker in C that records, categorizes, and summarizes transactions with secure storage using structured file I/O and core data structures.
- **Email Lite (Python, HTML, JavaScript, Cryptography):** Created a secure email client integrating a Python backend with an HTML/JS interface, featuring RSA key exchange, AES encryption, and SHA-256 hashing for message integrity.
- **Smart City Algorithm Engine (C, Algorithms):** Designed an algorithm-driven engine that optimizes smart-city resource allocation and urban infrastructure planning using classical graph and optimization algorithms.
- **Deep Fake Image Detection (AI/ML):** Developed a machine learning model capable of detecting manipulated or AI-generated images using deep learning-based feature extraction and classification techniques.

Technical Skills

- **Languages:** Python, C, C++
- **Backend:** Django, Node.js, Express.js
- **Frontend:** HTML, CSS
- **Clouds & Databases:** AWS, PostgreSQL
- **Web Technologies:** Docker, Socket.IO, Redis
- **Developer Tools:** VS Code, GitHub, CodeBlocks

Honours and Awards

- Solved 100+ Data Structures and Algorithms problems on LeetCode, developing strong proficiency in arrays, strings, hashing, recursion, linked lists, trees, and sorting algorithms, while demonstrating a solid understanding of time complexity and efficiently solving algorithmic tasks within optimal computational constraints.

- Practiced algorithmic problem solving consistently to improve coding efficiency and competitive programming ability.
- Strengthened algorithmic thinking and improved coding speed through structured DSA practice and contest participation.